

Test- III
Assignment
CS602/CO434 Image Processing

Marks: 20

Due date: 15/12/2020

1. (a) What is image filtering? 2
- (b) Consider the image $f(x, y)$ given below:

2	4	1	6	4
3	5	7	2	4
4	3	6	2	3
1	3	7	4	4

If the image is sharpened by applying the Sobel filter show the pixel value of the output sharpened image at location $f(2, 3)$ only. 3

2. (a) What is meant by a colour model? 1
- (b) Why is it necessary? 1
3. (a) What is the formal definition of image segmentation? 2
- (b) Provide a categorization of different segmentation methods. 1
4. How are isolated points and lines recognized? 2
5. Consider the one-dimensional image:
 $f(x)=[10\ 10\ 10\ 10\ 40\ 40\ 40\ 40\ 20\ 20]$.
 - (a) Compute the first and second order derivatives. 2
 - (b) Locate points of edges. 2
6. (a) What is a structuring element? 1
- (b) What is hit-or-miss transform? 1
- (c) The input binary image and structuring elements are shown below. Perform the erosion and dilation of the input image. The shaded 1 represent the center of the structuring element. 2

0	0	0	0	0	0	0
0	0	0	0	1	1	0
0	0	1	1	0	1	0
0	0	1	1	0	0	0
0	0	0	1	1	0	0
0	0	0	0	0	0	0

	1	
	1	
	1	1